PAnDA: Combating Negative Augmentation via Large Language Models for User Cold-Start Recommendations

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Abstract

The cold-start problem remains a long-standing challenge in recommender systems. Recent advances in large language models (LLMs) have opened new avenues for addressing cold-start scenarios through data augmentation. However, existing cold-start augmentation methods often suffer from *negative augmentation*, manifesting as *incomplete augmentation*, where generated interactions fail to comprehensively reflect user preferences, and *inaccurate augmentation*, where they conflict with user intent. These issues largely stem from two limitations: (1) the inability to effectively incorporate collaborative signals, which are critical for preference alignment, and (2) the lack of awareness of the downstream model's learning dynamics during data augmentation. To the best of our knowledge, the latter has not been studied in the literature.

Consequently, we propose a novel framework named PAnDA. To address the incomplete augmentation issue, we propose a model-agnostic preference-aligned augmentation module to iteratively extract and fuse textual information and collaborative information by user-user preference matching and user-item preference coherence, which together form a contextual cue to guide the augmentor to generate high-quality augmented data. To overcome the inaccurate augmentation issue, we propose a model-specific downstream-model-aware adaptation module to adaptively align the augmented data with the model's states during the training process, guided by gradient similarity. Extensive experiments on three public benchmark datasets demonstrate that PAnDA outperforms different groups of state-of-the-art cold-start recommendation methods in all scenarios. The source code is publicly available at https://github.com/YantongDU/PAnDA.

CCS Concepts

• Information systems → Personalization; **Information extraction**; **• Computing methodologies → Machine learning**.

CIKM '25, Seoul, Republic of Korea

ACM ISBN 979-8-4007-2040-6/2025/11
https://doi.org/10.1145/3746252.3761080

Keywords

Cold-start recommendations; large language models; data augmentation; meta-learning

ACM Reference Format:
Yantong Du, Rui Chen*, Xiangyu Zhao*, Qilong Han, and A. K. Qin. 2025. PAnDA: Combating Negative Augmentation via Large Language Models for User Cold-Start Recommendations. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25), November 10–14, 2025, Seoul, Republic of Korea.* ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3746252.3761080

1 Introduction

Recommender systems have played a crucial role in mitigating information overload in a wide range of real-world applications by efficiently providing online users with relevant content. Existing recommendation models, such as collaborative filtering [7] and content-based methods [4, 8], typically recommend appropriate items to users by learning user/item representations from their historical interactions (e.g., clicks, ratings, purchases). It is natural that this idea would fail in scenarios where user-item interactions are limited, which is known as the *cold-start* problem [2, 32], a long-standing challenge for recommender systems.

An intuitive strategy to address the cold-start problem is to generate additional user interactions (i.e., *data augmentation*) to enrich user behaviors and further guide model learning. This allows recommendation models to capture more diverse user preferences, as illustrated in Fig. 1(a). Some studies have explored multi-modal augmentation, leveraging auxiliary information such as images[38], audio [9], and text [30, 38] to simulate interactions that better represent users' interests. More recently, the emergence of large language models (LLMs) has opened up new opportunities for data augmentation in recommendation tasks [22, 30]. Owing to their extensive world knowledge and strong capabilities in language generation and reasoning, LLMs are increasingly regarded as promising augmentation tools in cold-start scenarios [35]. They can complement sparse user or item information and generate contextually appropriate augmented interactions. However, existing augmentation methods still suffer from significant limitations in cold-start settings. Due to the difficulty of multi-modal alignment and limited model generative capabilities, these methods struggle to accurately capture user preferences and can result in augmented interactions that contradict user intents, mislead model learning, and degrade recommendation performance, called *negative augmentation*, as shown in Fig. 1(b). It can be further categorized into two core challenges. From the data perspective, the reliance on multi-modal information often results in *incomplete augmentation*, as it fails to comprehensively capture user preferences. From the model perspective, limited generative capability can lead to *inaccurate augmentation*, where the augmented interactions misalign with user intents.

To achieve complete data augmentation, recent methods have attempted to integrate collaborative signals with multi-modal information [9, 31]. However, effectively fusing these heterogeneous sources remains an open challenge. This challenge is further exacerbated in cold-start scenarios, where limited interaction data makes it even more difficult to balance the complementary strengths of content semantics and collaborative patterns. As a result, augmented interactions are often biased toward a single modality or signal type, failing to provide a holistic representation of user preferences and ultimately degrading the performance of downstream recommendation models.

On the other hand, inaccurate augmentation arises from the capability limitations of generative models. For example, clickbait titles or mismatched image-text pairs [8, 31, 40] may guide the model to augment interactions that are superficially relevant but semantically inconsistent with the user's true intent. Incorporating such inaccurate augmentation into model training indiscriminately not only introduces label noise but can also distort learned preference distributions and increase overfitting risks. In extreme cases, this may cause the model to over-personalize or recommend irrelevant items, further deteriorating the user experience. Moreover, the augmented data will be consumed by a downstream recommendation model. Different models have disparate learning capabilities and thus expect different extents/types of data augmentation (e.g., the number of augmented interactions needed for a cold-start user), suggesting that LLM-based data augmentation needs to be aware of the downstream model. More specifically, the downstream model's training status needs to be considered in the data augmentation process. Therefore, mitigating inaccuracy while ensuring compatibility with the downstream recommendation model presents an additional challenge for LLM-based data augmentation.

To address these two challenges, we propose a novel preference-aligned and downstream-model-aware data augmentation framework PAnDA inspired by the pre-train and fine-tune paradigm [16, 17, 38], which consists of two complementary modules, as shown in Fig. 1(c). A *model-agnostic* preference-aligned augmentation module like a pre-training stage, and a *model-specific* downstream-model-aware adaptation module like a fine-tuning stage. Specifically, to mitigate incomplete augmentation, we focus on generating diverse and comprehensive user-item interactions. We perform the user-user preference matching by focusing on preference differences between users and designing a unique prompt to be used as a contextual cue to assist LLMs in generating augmented interactions. Additionally, we leverage user-item preference coherence to further model collaborative structures, enabling more personalised and accurate augmentation. These two components work in tandem to integrate both multi-modal content and collaborative signals, achieving more complete data augmentation. To address inaccurate augmentation, we further introduce a model-specific adaptation module. This component dynamically assesses the relevance of

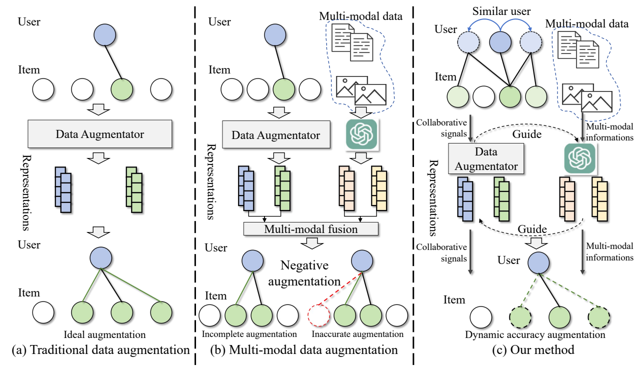

Figure 1: An illustration of different data augmentation methods in the cold-start recommendation scenario.

each augmented sample by monitoring the learning state of the downstream recommender. By selectively incorporating or discarding augmented interactions, it enhances the alignment between the augmented data and the model's learning objectives and prevents noisy or harmful samples from degrading performance.

To summarize, the main contributions of our work are as follows:

- We are the first to identify and study the **negative augmentation** in cold-start recommendation, highlighting that more augmented data does not necessarily lead to better performance. We reveal two underlying challenges: incomplete augmentation from the data perspective and inaccurate augmentation from the model perspective, which degrade the effectiveness of data augmentation methods.
- We introduce a novel preference-aligned and downstream-model-aware data augmentation framework PAnDA powered by LLMs. It consists of a *model-agnostic* preference-aligned augmentation module and a *model-specific* downstream-model-aware adaptation module, which together effectively address the two limitations. PAnDA also features a decoupled design to accommodate different combinations of LLMs and downstream recommendation models.
- We have performed extensive experiments on three real-world benchmark datasets and shown that PAnDA, being both preference-aligned and downstream-model-aware, can consistently outperform different groups of state-of-the-art cold-start recommendation methods in all scenarios.

2 Preliminary

In this section, we first introduce the problem formulation and notations, followed by a general overview of data augmentation in recommender systems.

Problem Formulation. Let $\mathcal{U}$ and $\mathcal{V}$ denote the sets of users and items, respectively. The user-item interaction matrix is defined as $\boldsymbol{A} \in 0, 1^{|\mathcal{U}|\times|\mathcal{V}|}$, where $A_{uv} = 1$ indicates that user u has interacted with item v. The interaction history of user u is denoted by $\mathcal{V}_u = v_1, v_2, \ldots, v_{|\mathcal{V}_u|}$. Collaborative filtering (CF) methods learn from $\boldsymbol{A}$ to obtain user and item ID-based embeddings $\boldsymbol{E} = \{\boldsymbol{E}_u, \boldsymbol{E}_v\}$ for prediction. However, such methods struggle in cold-start settings where user/item IDs are unseen. To address this, profile-based CF methods incorporate side information $\mathcal{P} = \{\mathcal{P}^U, \mathcal{P}^V\}$ for users and items, and learn representations using a function $f_{\Theta_{rec}}$ based on both $\boldsymbol{A}$ and $\mathcal{P}$. The model is trained by maximizing the posterior:

$$\Theta_{rec}^{*} = \arg\max_{\Theta_{rec}} p(\Theta_{rec}|\boldsymbol{A}, \mathcal{P}), \tag{1}$$

where $f_{\Theta_{rec}}$ will output the final user representation $\boldsymbol{h}_u$ contain both collaborative signals from $\boldsymbol{E}$ and side information p_u via:

$$\boldsymbol{h}_u = f_{\Theta_{rec}}(\boldsymbol{A}, p_u). \tag{2}$$

The item's final representation $\boldsymbol{h}_v$ can also be obtained similarly. **Data augmentation for Recommender Systems.** In cold-start scenarios, the sparsity of interactions motivates the use of data augmentation. Let $f_{\Theta_{aug}}$ denote the augmentation function, which generates synthetic interactions $\widetilde{\boldsymbol{A}}$ from $\boldsymbol{A}$ and $\mathcal{P}$ via:

$$\widetilde{\boldsymbol{A}} = f_{\Theta_{aug}}(\boldsymbol{A}, \mathcal{P}). \tag{3}$$

The downstream recommender model $f_{\Theta_{rec}}$ is then trained with the augmented data $\boldsymbol{A}_{aug} = \{\boldsymbol{A}, \widetilde{\boldsymbol{A}}\}$ to explore user preferences, which can be expressed via:

$$\Theta^{*} = \arg\max_{\Theta} p(\boldsymbol{A}_{aug}, \mathcal{P}), \tag{4}$$

where Θ is the trainable parameter of the models, $\Theta = \{\Theta_{aug}, \Theta_{rec}\}$. After training with augmented data $\boldsymbol{A}_{aug}$, the recommender model $f_{\Theta_{rec}}$ is used to predict preference score $\hat{y}_{u,v}$ by ranking the likelihood of user u will interact with item v.

3 Methodology

To address the user cold-start problem, we propose the PAnDA framework, illustrated in Figure 2. First, we introduce a model-agnostic *preference-aligned augmentation* module. It leverages textual and collaborative signals for *user-user preference matching*, where LLMs serve as the Textual Information Augmentor (TIA). Additionally, *user-item preference coherence* is used to capture user interests and item features, with LLMs enhancing personalization by guiding the integration of collaborative signals and mitigating incomplete augmentation. Second, we present a model-specific *downstream-model-aware adaptation* module. This component aligns augmented data with the training signals of downstream recommenders, enabling effective preference-aligned augmentation and alleviating data sparsity in cold-start scenarios.

3.1 Model-Agnostic Preference-Aligned Augmentation

3.1.1 User-User Preference Matching. To address the cold-start problem and effectively leverage auxiliary information for interpreting user preferences and item characteristics, we focus on textual signals and employ LLMs as the Textual Information Augmentor (TIA). By converting the augmentation task into a natural language description, LLMs generate meaningful user-item interaction pairs for each user u, drawing on their strong reasoning capabilities and broad knowledge. We also incorporate interaction histories from similar users as references to enhance augmentation quality.

Specifically, for each user u, we construct a similar user set $\boldsymbol{S}_u$. In cold-start scenarios, embedding-based similarity is often unreliable due to sparse interactions. Instead, we measure similarity based on interaction history. Let $\boldsymbol{A}_u = \{0, 1\} \in \mathbb{R}^{1\times|V|}$ denote the binary interaction vector of user u, we compute similarity with user m as:

$$s_{u,m} = \boldsymbol{A}_u \odot \boldsymbol{A}_m, \tag{5}$$

where $\odot$ denotes element-wise multiplication. We select the top-K similar users $\boldsymbol{S}_u$ as auxiliary context and combine each user's profile (e.g., age, gender), interaction history, and candidate item set $\boldsymbol{C}_u$ to construct a textual prompt $\mathbb{P}_u$ for the LLM. The LLM then generates an augmented interaction pair for user u, consisting of a preferred item $v_u^{+,t}$ and a non-preferred item $v_u^{-,t}$ from $\boldsymbol{C}_u$ via:

$$\begin{aligned} \mathbb{P}_u &= \mathrm{Text}(u, \boldsymbol{S}_u, \boldsymbol{C}_u, \boldsymbol{A}, \mathcal{P}), \\ \{v_u^{+,t}, v_u^{-,t}\} &= \mathrm{LLM}(\mathbb{P}_u), \end{aligned} \tag{6}$$

where $\mathrm{Text}(\cdot)$ denotes the prompt construction function. The candidate set $\boldsymbol{C}_u$ is obtained by hard sampling high-ranking items from a base recommender (e.g., BPR [21], LightGCN [6]). This process yields the positive and negative augmented interaction sets $\{\widetilde{\mathcal{V}}_u^{+,t}\}_{u\in\mathcal{U}}$ and $\{\widetilde{\mathcal{V}}_u^{-,t}\}_{u\in\mathcal{U}}$, where $|\widetilde{\mathcal{V}}_u^{+,t}| = |\widetilde{\mathcal{V}}_u^{-,t}| = M$ for each user u. By constraining augmentation to a pre-filtered candidate set and incorporating similar users' interactions as context, we enhance generation accuracy and mitigate noise from sparse data. Considering token-length limits of LLMs [3, 43], we avoid feeding the full item set and instead rely on a compact, informative subset. Overall, the proposed *Textual Interaction Augmentation (TIA)* module enables text-driven augmentation guided by collaborative signals, balancing interpretability and personalization.

3.1.2 User-Item Preference Coherence. Although LLMs, as the TIA, fully leverage auxiliary information, they have a critical limitation: the upper bound of the augmented data quality depends on the candidate item set C_u. Unfortunately, due to the constraints of the cold-start scenario, the base recommender also struggles to accurately capture user preferences, leading to varying quality for candidate sets. Additionally, LLMs primarily rely on processing and understanding input text, which leads to the fact that the augmented data generated by LLMs still has limitations. Due to the input token limitations of LLMs and the difficulty of incorporating collaborative signals from interaction data into LLMs and further gaining attention, we generate augmented data complemented by collaborative signals and propose the Collaborative Signal Augmentor (CSA).

3.1.3 Meta Masked Autoencoder (MetaMAE). To leverage collaborative information, we fine-tune the pre-trained model $f_{\Theta_{rec}^{pt}}$. For user u and augmented interacted item set $\mathcal{V}_{u,t} = \mathcal{V}_u \cup \widetilde{\mathcal{V}}_u^{+,t}$, we can obtain the item set representations via:

$$\boldsymbol{H}_u = Stack(\{\boldsymbol{h}_i\}_{i\in\mathcal{V}_{u,t}}), \tag{7}$$

which $Stack(\cdot)$ is the vector stacking operation. $\boldsymbol{h}_i \in \mathbb{R}^d$ is the representation of item i and obtained through Eq. (2), and $\boldsymbol{H}_u \in \mathbb{R}^{|\mathcal{V}_{u,t}|\times d}$. To learn an accurate and comprehensive representation of the user, we utilize the augmented interactions generated by LLMs from the perspective of items to guide the model's learning, thereby incorporating rich textual information into the item representations. To mitigate the impact of incomplete augmentation, we employ a Masked Autoencoder (MAE) to enhance the user/item representations of users and items.

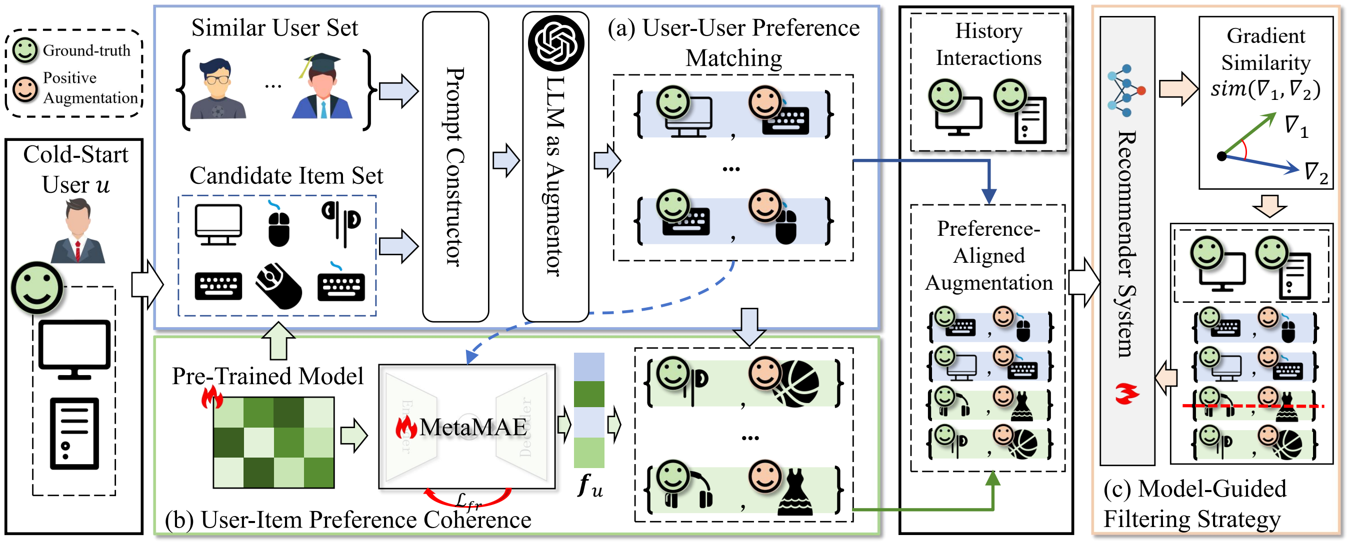

Figure 2: The architecture of the proposed PAnDA model. (a) demonstrates the textual information data augmentation process with LLMs. (b) describes the MetaMAE augment data with the collaborative signals. (c) introduces the downstream-model-aware filtering strategy for filtering model-mismatched interactions.

Task Description

```
You are a movie recommendation system and required to
recommend user A who is a [age]-year-old [gender]
person and the occupation is [occupation] with movies
based on user history that each movie with title, year,
type. For reference, we also will list a similar user
B's history.
```

Prompt

```
User A's history:
[movie title], [release year], [type]
...
[movie title], [release year], [type]
User B's history:
[movie title], [release year], [type]
...
[movie title], [release year], [type]
Candidates:
[id] [movie title], [release year], [type]
...
[id] [movie title], [release year], [type]
Please output the index of user A's favorite and least
favorite movie. Please give the index in [] from
candidates.
```

(a) The structure of the LLM augmentation task

```
You are a movie recommendation system and
required to recommend user A who is a 25-
year-old male person and the occupation is
writer with movies based on user history
that each movie with title, year, type.
For reference, we also will list a similar
user B's history.
User A's history:
Butch Cassidy and the Sundance Kid, 1969,
Action Comedy Western
Home Alone, 1990, Children's Comedy
User B's history:
Sabrina, 1995, Comedy Romance
Twelve Monkeys, 1995, Drama Sci-Fi
While You Were Sleeping, 1995, Comedy
Romance
Candidates:
[204] Back to the Future, 1985, Comedy
Sci-Fi
[98] Silence of the Lambs, The, 1991,
Drama Thriller
[50] Star Wars, 1977, Action Adventure
Romance Sci-Fi War
Please output the index of user A's
favorite and least favorite movie. Please
give the index in [] from candidates.
```

```
204 98
```

(b) Example of LLM aumentation

Figure 3: An illustration of the structure of prompts. The figure shows the prompt designed for movie datasets. For the Book-Crossing dataset, we use *[id], [book title], [author], [genre]* as descriptors.

First, for user u and the set of positive samples $\mathcal{V}_u^+$ generated by LLMs, we select a subset $\widetilde{\mathbf{v}}_u \subseteq \widetilde{\mathcal{V}}_u^{+,t}$ and mask their representations with a mask token $[MASK]$, represented as $\boldsymbol{h}_{[MASK]}$ (e.g., a learnable vector or mean pooling). The masking operation via:

$$\widetilde{\boldsymbol{h}}_v = \begin{cases} \boldsymbol{h}_v & \text{if } v \notin \widetilde{\mathbf{v}}_u \\ \boldsymbol{h}_{[\mathrm{MASK}]} & \text{if } v \in \widetilde{\mathbf{v}}_u. \end{cases} \tag{8}$$

It is worth noting that we only perform the mask operation on the augmented interactions to make the model more robust without losing the original information.

Second, we use the masked user interaction set $\mathcal{V}_u^M$ as the input to the autoencoder and reconstruct the representations via:

$$\widehat{\boldsymbol{H}}_u, \boldsymbol{z}_u = AutoEncoder(\boldsymbol{H}_u^M; \theta_{AE}), \tag{9}$$

where $\boldsymbol{z}_u$ represents the latent representation of user u. θ_{AE} is the trainable parameter of the autoencoder.

Third, we attempt to reconstruct the representation of the interacted item set $\mathcal{V}_{u,t}$ for user u. To enhance the representational capability of items, we use a feature restoration loss via:

$$\mathcal{L}_{fr}^u = \frac{1}{|\mathcal{V}_{u,t}|} \sum_{v \in \mathcal{V}_{u,t}} \left(1 - \frac{\widehat{\boldsymbol{H}}_u \cdot \boldsymbol{H}_u}{\left\|\widehat{\boldsymbol{H}}_u\right\| \cdot \|\boldsymbol{H}_u\|} \right)^\gamma, \tag{10}$$

where γ is the scaling factor, which is a hyperparameter.

Last, we aggregate the latent representation of user u that incorporates item-level collaborative signals and textual information, $\boldsymbol{z}_u$, with the profile-based user representation, $\boldsymbol{h}_u$ obtained through Eq. (2), to conduct the user's final collaborative representation via:

$$\boldsymbol{f}_u = (1 - \alpha)\boldsymbol{h}_u + \alpha \boldsymbol{z}_u, \tag{11}$$

where α is the trainable parameter. Then, we can conduct the prediction score of the user u to the item v via:

$$\hat{y}_{u,v}^c = \boldsymbol{f}_u \cdot \boldsymbol{h}_v. \tag{12}$$

Subsequently, we select the top-M items with the highest prediction scores as augmented positive samples $\mathcal{V}_u^{+,c}$. Conversely, we select the bottom-M items with the lowest scores as negative samples $\mathcal{V}_u^{-,c}$. The collaborative signal will then be fed into the next iteration to refine the candidate item set $\boldsymbol{C}_u$.

3.1.4 Meta optimization for MetaMAE. Since each user's preferences are different, using a shared-parameter autoencoder would struggle to capture the personalized differences among users accurately. In cold-start scenarios, the limited interaction data can also lead to undifferentiated representations of users or items. Inspired by meta-learning [33, 42], we designed a meta-optimization strategy to ensure each user has a personalized autoencoder that captures their unique preferences via:

$$\begin{aligned} &\min_{\Theta} \sum_{u \in \mathcal{U}} \mathcal{L}_{rec}(\mathcal{V}_u, \widetilde{\mathcal{V}}_u^t, \widetilde{\mathcal{V}}_u^c(\theta_{AE}^{u,*}); \Theta), \\ &s.t., \theta_{AE}^{u,*} \leftarrow \arg\min_{\theta_{AE}} \mathcal{L}_{fr}(\mathcal{V}_u, \widetilde{\mathcal{V}}_u^t, \Theta_{rec}^{pt}; \theta_{AE}), \end{aligned} \tag{13}$$

where $\theta_{AE}^{u,*}$ represents the personalized autoencoder parameters for user u after convergence through training with the augmented interaction data. V_u denotes the original set of interacted items for the user u, while $\widetilde{\mathcal{V}}_u^t = \{\widetilde{\mathcal{V}}_u^{+,t}, \widetilde{\mathcal{V}}_u^{-,t}\}$ represents the item pairs augmented by the TIA. $\widetilde{\mathcal{V}}_u^c(\theta_{AE}^{u,*}) = \{\widetilde{\mathcal{V}}_u^{+,c}, \widetilde{\mathcal{V}}_u^{-,c}\}$ denotes the item pairs augmented by the CSA using parameters $\theta_{AE}^{u,*}$. $\mathcal{L}_{rec}$ is the recommendation loss of the downstream recommender model f_Θ, which will be elaborated later. To fully integrate with the training process of the downstream recommender system, we adopted the end-to-end optimization strategy. Therefore, we employed the reparameterization trick [15] to implement the data augmentation process of the CSA. MetaMAE incorporates textual information to enhance the representation capability of the model, learning comprehensive user/item representations that integrate both collaborative signals and textual information. Additionally, using a bi-level meta-optimization strategy distinguishes between different users when generating augmented data, thereby producing comprehensive and personalized augmented data. This approach effectively addresses the issue of incomplete augmentation.

3.2 Model-Specific Downstream-Model-Aware Adaptation

We obtain augmented interactions for user u, $\widetilde{\mathcal{V}}_u = \widetilde{\mathcal{V}}_u^t, \widetilde{\mathcal{V}}_u^c$, including true positives and preference-aligned samples. However, not all of these interactions are equally useful for the downstream recommender. In cold-start settings, the quality of augmented data varies, and limited user understanding may cause some samples to diverge from the model's current learning trajectory. Training on all samples indiscriminately risks noise, misleading optimization, and performance drops. Since models differ in training dynamics, they may react differently to the same data. Thus, it is essential to check each interaction's compatibility with the model's current state. Inspired by curriculum learning, we evaluate sample–model alignment at each iteration, enabling the model to emphasize informative data and filter mismatched ones, improving representations without distorting user intent.

Specifically, we define the training loss for user u's interactions:

$$\mathcal{L}(\mathcal{V}_u) = \sum_{v^+,v^- \in \mathcal{V}_u} \mathcal{L}_{rec}(u, v^+, v^-), \tag{14}$$

where $\mathcal{L}_{rec}(\cdot)$ is the loss function of the downstream recommender parameterized by Θ. Some prior work filters out augmented interactions with high loss, assuming they are model-mismatched. However, this overlooks a key limitation: high loss does not necessarily imply low quality. In many cases, such interactions are informative hard examples that can improve model robustness. Discarding them may lead to underfitting and missed learning opportunities. Conversely, low-loss interactions may be uninformative or even misleading if they poorly reflect user preferences. Therefore, loss alone is not a reliable signal for evaluating augmentation quality. Instead, we propose to use gradient signals [1, 11] to assess the alignment between each augmented interaction and the model's current learning direction. By measuring the gradient similarity between augmented and original interactions, we can more accurately identify and retain useful samples while filtering out those inconsistent with the model's optimization path.

First, we compute the average gradient of the loss over the original interaction set $\mathcal{V}_u$, which represents the model's direction for updating parameters based on the user's actual preferences via:

$$\nabla_\Theta \mathcal{L}(\mathcal{V}_u) = \frac{1}{|\mathcal{V}_u|} \sum_{\{v^+,v^-\} \in \mathcal{V}_u} \nabla_\Theta \mathcal{L}_{rec}(u, v^+, v^-), \tag{15}$$

Second, for each augmented interaction pair $\tilde{v}^+, \tilde{v}^-$, we calculate the cosine similarity between its gradient and the gradient of the original user interactions to evaluate alignment via:

$$sim(\{\tilde{v}^+, \tilde{v}^-\}, \mathcal{V}_u) = \left\langle \nabla_\Theta \mathcal{L}_{rec}(u, \tilde{v}^+, \tilde{v}^-), \nabla_\Theta \mathcal{L}(\mathcal{V}_u) \right\rangle, \tag{16}$$

where $\langle \cdot, \cdot \rangle$ denotes a cosine similarity operator between gradients.

Last, for each user u, we sort all augmented interaction pairs $\{\tilde{v}^+, \tilde{v}^-\} \in \widetilde{\mathcal{V}}_u$ by their similarity scores and discard those with the lowest alignment. The model then updates its parameters Θ using the remaining interactions, ensuring that training is guided by interactions consistent with the user's original preference signals.

3.3 Model Optimization

After obtaining the comprehensive augmented interactions $\widetilde{\mathcal{V}}u$ from TIA and CSA, we use them to train a new recommender model f_Θ. The goal is to address the cold-start challenge by leveraging high-quality augmented data to learn accurate and expressive user/item representations, thereby enhancing recommendation performance. To better capture the underlying relationships from limited interactions, we adopt Bayesian Personalized Ranking (BPR) as the training objective via:

$$\mathcal{L}_{rec}(u, v^+, v^-) = -\log(\sigma(\hat{y}_{u,v^+} - \hat{y}_{u,v^-})), \tag{17}$$

where each training triplet (u, v^+, v^-) is sampled from the union of the user's historical and augmented interactions, i.e., $\mathcal{V}u \cup \widetilde{\mathcal{V}}u$. The predicted scores $\hat{y}_{u,v^+}$ and $\hat{y}_{u,v^-}$ are generated by f_Θ.

The entire model adopts a bi-level optimization end-to-end training strategy, where the training parameters include the parameters θ_{AE} of the MetaMAE and the parameters Θ of the downstream recommender model. The objective is shown in Eq. (13). Similar to the training approach in meta-learning, the model training primarily consists of inner-loop optimization and outer-loop optimization.

Inner-Loop Optimization. The primary goal of this optimization is to obtain the user-specific personalized autoencoder $\theta_{AE}^{u,*}$ for user u through rapid gradient descent, thereby obtaining a comprehensive and accurate representation $\boldsymbol{f}_u$ as shown in Eq. (11). Then, we can generate augmented interactions $\mathcal{V}_u^c$ that primarily contain collaborative signals supplemented by textual signals, as shown in Eq. (12). As advised by [19], we use one gradient descent to approximate the final optimized result via:

$$\theta_{AE}^{u,*} \approx \theta_{AE} - \omega_1 \nabla_{\theta_{AE}} \mathcal{L}_{fr}^u, \tag{18}$$

where ω_1 is the learning rate of inner-loop optimization.

Outer-Loop Optimization. The optimization objective of this optimization, as shown in Eq. (13), remains to enhance the final recommendation performance of recommender system f_Θ. Additionally, to fully utilize the augmented data obtained from textual information and collaborative signals, we adaptively filter out augmented interactions that do not match the model with the help of the training signals. This approach helps the model learn more accurate user/item representations without altering the user's original intent. Ultimately, we update the parameters θ_{AE} of the MetaMAE and the parameters Θ of the downstream recommender model via:

$$\theta_{AE} = \theta_{AE} - \omega_2 \sum_{u \in \mathcal{U}} \nabla_{\theta_{AE}} \mathcal{L}(\mathcal{V}_u, \widetilde{\mathcal{V}}_u^t, \widetilde{\mathcal{V}}_u^c(\theta_{AE}^{u,*})),$$
$$\Theta = \Theta - \omega_3 \sum_{u \in \mathcal{U}} \nabla_{\Theta} \mathcal{L}(\mathcal{V}_u), \tag{19}$$

where ω_2, ω_3 are the learning rates of outer-loop optimization. It is worth mentioning that we focus on updating the meta-model parameters θ_{AE} rather than the user-specific personalized autoencoder $\theta_{AE}^{u,*}$ obtained in the inner-loop optimization.

4 Experiments

In this section, we conduct experiments to answer the following research questions (RQs):

- **RQ1**: How does our PAnDA perform in the cold-start scenario compared to the current state-of-the-art baselines?
- **RQ2**: What is the impact of critical components on the performance?
- **RQ3**: How do different LLMs impact PAnDA?
- **RQ4**: How sensitive is the model to different parameters?
- **RQ5**: How does the model augmented sample differ from other samples?

4.1 Experimental Setup

4.1.1 Datasets. We evaluate PAnDA on three widely used real-world benchmark datasets: (1) **MovieLens (ML-1M)**[1], (2) **Netflix**[2], (3) **Book-Crossing**[3].

Following prior works [13, 14, 18], we simulate cold-start scenarios by retaining only users with no more than 100 interactions. Each dataset is split into training, validation, and test sets with a ratio of 8:1:1. Dataset statistics are summarized in Table 1.

4.1.2 Evaluation Metrics. We assess model performance using: (1) *Recall@K* (R@K), (2) *Normalized Discounted Cumulative Gain* (N@K), and (3) *Precision@K* (P@K). To mitigate test sampling bias, we adopt the all-ranking evaluation strategy [31]. Results are averaged over five independent runs, with K set to 10, 20, and 50. Statistical significance is assessed via p-values computed against the best-performing baseline.

4.1.3 Competing models. To demonstrate the effectiveness of PAnDA, we compare our model with: (i) CF-based methods including **BPR** [21], **LightGCN** [6] and **NGCF** [28]. (ii) Augmentation-based methods including **DropoutNet** [24], **CL4SRec** [36], **L2Aug** [25], **KAR** [34] and **LLMRec** [30]. (iii) Cold-start methods including **M2EU** [33] and **TDAS** [42].

4.1.4 Implementation details. We fix the embedding size of each profile feature (e.g., age, gender) to 32 and set the training batch size to 2048 for both datasets. Embedding parameters are initialized using a Gaussian distribution [23]. We apply early stopping when N@50 does not improve for more than 10 consecutive iterations, selecting the best-performing model during training as the final one. For all baseline models, hyperparameters are either set according to their original papers or carefully tuned on the validation set, with the best results reported. We choose the temperature γ from 0, 0.6, 0.8, 1, and initialize the aggregation parameter α to 0.01. The learning rates $\omega_1, \omega_2, \omega_3$ are searched in the ranges $[5e^{-5}, 1e^{-3}]$, $[1e^{-4}, 8e^{-4}]$, and $[1e^{-4}, 8e^{-4}]$, respectively. We set the number of candidate items to 20 for all datasets. Each user receives 5 augmented positive and 5 negative samples, and 3 item pairs are filtered out using a downstream-model-aware strategy. The LLM used in PAnDA (Table 2) is GPT-4o[4], while KAR and LLMRec use LLaMA3-8B-Chat[5] due to cost considerations. We also report PAnDA's performance with LLaMA3-8B-Chat for comparison. Our implementation is based on PyTorch 2.0.0 and Python 3.11.1, with the RecBole library [41]. Experiments are run on a workstation with an Intel Xeon Platinum 2.40GHz CPU, NVIDIA Quadro RTX 8000 GPU, and 754GB RAM.

4.2 Overall Performance Comparison (RQ1)

We report the main experimental results in Table 2. From the results, we can draw the following conclusions:

First, our model, PAnDA, consistently outperforms all other baseline models, indicating its robustness and effectiveness in addressing the cold-start. This superiority is achieved by incorporating a high-quality data augmentation strategy that generates comprehensive augmented data and adaptively selects high-quality augmented samples based on model training signals. This adaptive approach ensures that the model benefits from the most relevant and informative data, leading to significant performance gains.

Second, PAnDA demonstrates substantial improvements over traditional cold-start recommender models such as DropoutNet and state-of-the-art MAML-based methods. This result underscores the importance of leveraging textual signals in cold-start scenarios, as these signals provide crucial contextual information that needs to be included in sparse user-item interaction data. Moreover, the results highlight that data augmentation, particularly when combined with adaptive selection mechanisms, offers a promising and practical direction for overcoming the limitations of cold-start scenarios.

At last, PAnDA also demonstrates substantial improvements over traditional cold-start recommendation methods, such as DropoutNet, and state-of-the-art MAML-based methods, such as TDAS and M2EU. These textual signals provide essential contextual information that complements collaborative signals, particularly in cold-start scenarios where interaction data is limited. Furthermore, the results highlight the critical role of adaptive data selection mechanisms. By dynamically filtering and selecting the most relevant augmented samples, PAnDA ensures that the model learns from preference-aligned, contextually aligned data, setting a new benchmark for addressing cold-start challenges in recommendation systems. In summary, the experimental results validate the effectiveness of PAnDA in overcoming the limitations of existing cold-start recommendation methods. By integrating textual and collaborative signals into a unified augmentation framework and employing an adaptive filtering strategy, PAnDA delivers state-of-the-art performance across multiple datasets and metrics, establishing a robust and scalable solution for cold-start scenarios.

[1]https://movielens.org/
[2]https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data
[3]http://www2.informatik.uni-freiburg.de/~cziegler/BX/
[4]https://platform.openai.com/
[5]https://llama.meta.com/

Table 1: Statistics of the experimental datasets

Statistics	ML-1M	Netflix	Book-Crossing
#User	3,132	245,281	103,459
#Item	3,354	17,761	189,284
#Interaction	156,507	10,627,773	493,175
Sparsity	98.5101%	99.7560%	99.9975%
Avg. #interactions per user	49.9862	43.3291	4.7669
user profiles	age, gender, occupation, zip_code	age, gender, occupation, zip_code	location, age
item profiles	movie_title, release_year, class	movie_title, release_year, class	book_title, book_author, publication_year, publisher, genre
Range of ratings	1~5	1~5	1~10

Table 2: The experimental comparison between PAnDA and the SOTA cold-start methods on the two benchmark datasets. The best results are marked in bold, and the second-best results are underlined. All improvements are significant under a two-sided t-test with $p < 0.05$ over the best baselines.

Datasets	Metrics	CF-based methods			Augmentation-based Methods					Cold-start Methods		PAnDA	Improv.
		BPR	LightGCN	NGCF	DropoutNet	CL4SRec	L2Aug	KAR (LLM-based)	LLMRec (LLM-based)	TDAS	M2EU		
ML-1M	R@10	0.2049	0.2081	0.18	0.2719	0.1478	0.2641	0.5084	0.5106	0.4591	0.4813	**0.5891**	15.37%
	N@10	0.1708	0.1727	0.1505	0.2584	0.0767	0.1343	0.5148	0.4845	0.4266	0.4548	**0.5643**	16.47%
	R@20	0.3015	0.3082	0.2717	0.3689	0.1837	0.3804	0.5933	0.6184	0.5315	0.5798	**0.6997**	13.15%
	N@20	0.2099	0.2127	0.187	0.2912	0.0873	0.1634	0.5367	0.5087	0.4598	0.4757	**0.6017**	18.28%
	R@50	0.4707	0.4752	0.4331	0.5165	0.2845	0.5506	0.7415	0.7612	0.6847	0.7214	**0.8516**	11.88%
	N@50	0.2638	0.266	0.2376	0.3284	0.1204	0.1972	0.6067	0.5536	0.2964	0.5249	**0.6644**	20.01%
Book -Crossing	R@10	0.015	0.0158	0.0088	0.0164	0.0115	0.0202	0.0229	0.0229	0.0281	0.0197	**0.0315**	37.55%
	N@10	0.0082	0.0091	0.0066	0.0092	0.0059	0.0098	0.0101	0.0115	0.0151	0.0104	**0.0161**	40.00%
	R@20	0.0243	0.0259	0.0159	0.0268	0.0154	0.0324	0.0331	0.0349	0.0382	0.0307	**0.0486**	39.26%
	N@20	0.0107	0.0118	0.0079	0.0121	0.0072	0.0143	0.0150	0.0157	0.0191	0.0144	**0.0233**	48.41%
	R@50	0.0419	0.0466	0.0256	0.0471	0.0292	0.0564	0.0607	0.0631	0.0701	0.0568	**0.0811**	28.53%
	N@50	0.0144	0.0162	0.0091	0.0171	0.0103	0.0203	0.0263	0.0278	0.0324	0.0204	**0.0351**	26.26%
Netflix	R@10	0.045	0.0458	0.0388	0.0464	0.0207	0.0497	0.0529	0.0529	0.0482	0.0497	**0.0615**	16.26%
	N@10	0.0382	0.0391	0.0352	0.0392	0.0115	0.0213	0.0415	0.0415	0.0397	0.0404	**0.0461**	11.08%
	R@20	0.0543	0.0559	0.0414	0.0568	0.0326	0.0634	0.0639	0.0649	0.0581	0.0607	**0.0786**	21.11%
	N@20	0.0407	0.0418	0.0366	0.0421	0.0143	0.0272	0.0394	0.0457	0.0424	0.0444	**0.0533**	16.63%
	R@50	0.0719	0.0766	0.0556	0.0771	0.0482	0.0861	0.0906	0.0931	0.0781	0.0868	**0.1111**	19.33%
	N@50	0.0444	0.0462	0.0391	0.0471	0.0176	0.0314	0.0523	0.0578	0.0484	0.0504	**0.0651**	12.63%

Table 3: Ablation study on ML-1M. LLMRec emerges as the second-best performing baseline overall.

Variants	N@10	R@20	N@20	R@50	N@50
LLMRec	0.4845	0.6184	0.5087	0.7612	0.5536
w/o TIA	0.4415	0.5626	0.4703	0.7056	0.5184
w/o CSA	0.5007	0.6176	0.5174	0.7749	0.5689
w/o DFS	0.4688	0.5991	0.4954	0.7167	0.5411
PAnDA	**0.5643**	**0.6997**	**0.6017**	**0.8516**	**0.6644**

4.3 Ablation Study (RQ2)

We conducted a series of ablation experiments on ML-1M to investigate the contribution of components applied within PAnDA, as shown in Table 3. **w/o TIA**: The removal of TIA leads to a substantial drop in performance across all metrics. This is primarily because, without integrating textual signals, the augmented samples generated lack sufficient contextual richness, thereby increasing the prevalence of false positive samples. These results underscore the importance of textual information in enhancing the model's understanding of user preferences. The ability of LLMs to process and integrate these textual signals plays a pivotal role in improving the overall quality of the augmented data. **w/o CSA**: The exclusion of CSA also results in noticeable performance degradation.

Table 4: Ablation study on ML-1M. Impact of different LLMs on accuracy, cost, and latency

LLMs	R@20	N@20	Cost	Latency
LLama2-7B-chat	0.6178	0.4981	-	33s
LLama3-8B-chat	0.6813	0.5847	-	1.5s
gpt-3.5-turbo	0.6345	0.5381	$24.2	27s
gpt-4o-mini	0.6807	0.5833	$2.4	13s
gpt-4o	**0.6997**	**0.6017**	$40.3	18s

Table 5: Analysis of $|C|$ on ML-1M and Book-Crossing.

$\|C\|$	ML-1M			Book-Crossing		
	R@20	N@20	P@20	R@20	N@20	P@20
3	0.6514	0.5814	0.1849	0.0407	0.0204	0.0048
10	0.6866	0.5948	0.1906	0.0467	0.0228	0.0051
20	**0.6997**	**0.6017**	**0.1987**	**0.0486**	**0.0233**	**0.0053**
30	0.6915	0.6003	0.1954	0.0479	0.0231	0.0052

Without CSA, the model relies solely on LLM-generated augmentations, which often produce partially aligned samples that fail to capture user preferences. This incomplete alignment hampers the downstream recommender model's ability to accurately and comprehensively learn user/item representations, highlighting collaborative signals' crucial role in ensuring the augmented data's robustness and accuracy. **w/o DFS**: The absence of downstream-model training feedback signals leads to a decline in performance, as it hinders the model's ability to filter out samples mismatched with the model's current training trajectory. Without this pruning, the model struggles to adapt to the diverse set of augmented samples, resulting in less consistent user/item representations and ultimately leading to suboptimal recommendation performance. This emphasizes the necessity of integrating feedback signals to maintain the relevance and quality of the training data.

4.4 LLMs Analysis (RQ3)

Table 4 presents the results of the ablation study analyzing the impact of different LLMs as textual information augmentors on PAnDA. We include both open-source models (LLaMA series) and closed-source models (ChatGPT series), and compare their effects on accuracy, cost, and latency. The results show that the choice of LLM has a substantial influence on performance: stronger models such as GPT-4o achieve the best Recall@20 and NDCG@20, while LLaMA3-8B-chat also delivers competitive accuracy at negligible cost and extremely low latency when deployed on optimized hardware. This indicates a clear correlation between model scale and performance, with larger and more advanced LLMs providing more accurate recommendations. Overall, the findings demonstrate that employing higher-capacity LLMs can significantly improve PAnDA, highlighting the importance of carefully selecting models that balance accuracy gains with efficiency and cost considerations.

4.5 Hyperparameter Analysis (RQ4)

Analysis of $|C|$. Since the input token constraints of the LLMs, coupled with the problem of false positive augmented samples, we use the candidate item set C to limit the candidate items based on the LLMs augmented samples. Due to cost constraints, we explored

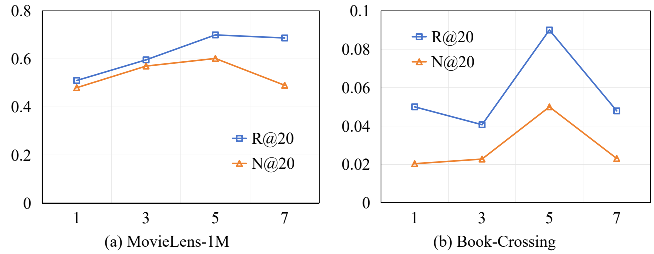

Figure 4: Performance comparison (Recall@20 and NDCG@20) with varying numbers of augmented data pairs.

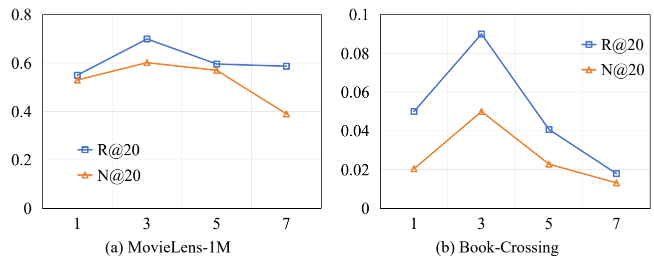

Figure 5: Performance comparison (Recall@20 and NDCG@20) with discard augmented data pairs.

$\{3, 10, 20, 30\}$, and Table 5 shows that $|C| = 20$ gives the best results. Smaller values limit the choices, while larger values make the recommendation more difficult.

Analysis of the #.augmented data pairs. We can observe from Figure 4 that the impact of the number of augmented sample pairs varies across different datasets. Unlike ML-1M, the Book-Crossing dataset is sparse, making it difficult to generate comprehensive and accurate augmented samples. As a result, the model is more sensitive to the number of augmented samples. This also indicates that the quality of generated augmented samples is critically important.

Analysis of the #.discard augmented data pairs. We can observe from Figure 5 that there are model mismatched augmented samples. The model's performance improves by discarding these samples based on the model's training signals. However, excessive discarding may lead to the loss of highly informative and high-quality augmented data, harming the model's performance.

Analysis of the #.similar user. We can observe from Figure 6 that although the introduction of similar users can improve LLM's ability to understand user preferences, over-information brings performance degradation because there is too much textual information, and it is difficult for LLM to focus on the key information.

4.6 Case Study (RQ5)

As shown in Figure 7, the differences in augmented data generated by various methods for cold-start tasks are evident. The left side shows the ground truth user interaction data and the right side displays the augmented data distribution. We assess the quality of augmented samples using the maximum cosine similarity, denoted as q, between each augmented sample's embedding and the ground truth samples. This metric indicates the quality of the augmented samples. Traditional methods like L2Aug struggle with accurately capturing user preferences, leading to many false-positive samples ($q \leq 0.25$), which negatively impact model learning. LLM-based

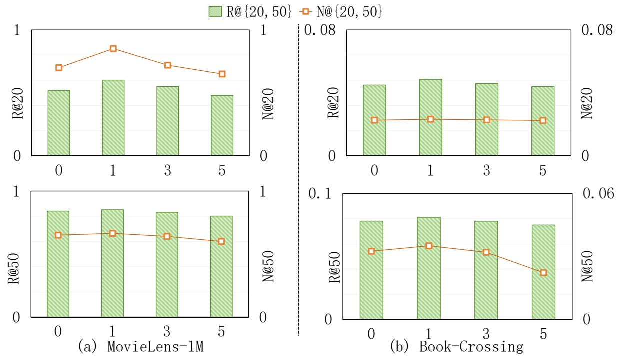

Figure 6: Analysis of the #similar users on ML-1M and Book-Crossing.

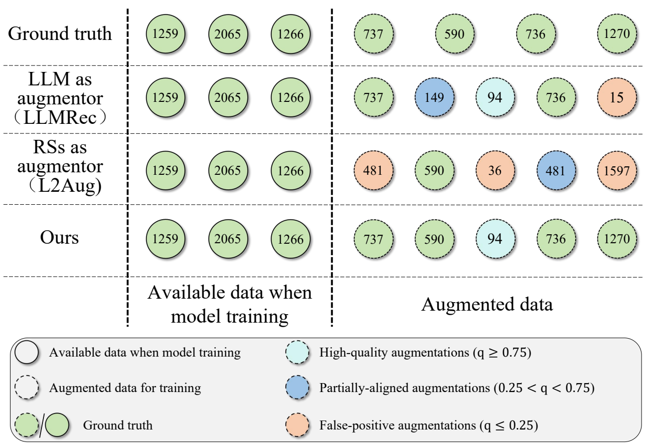

Figure 7: Case study on augmentation samples by different augmentors.

methods (LLMRec) use world knowledge to generate preference-aligned samples but still produce partially aligned and false-positive samples, showing the limitations of relying solely on textual information. In contrast, PAnDA generates augmented data that closely aligns with valid user preferences, effectively capturing accurate preferences and eliminating false-positive and partially aligned samples. By combining textual augmentation, collaborative signals, and a downstream-model-aware filtering strategy, PAnDA addresses data sparsity and quality issues in cold-start scenarios, providing high-quality, preference-aligned augmented samples.

5 Related Work

Cold-start Recommendation. To address this issue, many works use auxiliary information to improve cold-start user or item representations, such as social networks [16] or cross-domain data [4]. Graph Neural Networks (GNNs) further capture high-order semantics from knowledge graphs [5] and heterogeneous networks [37]. When side information is limited, contrastive learning [27] helps refine collaborative embeddings. More recently, meta-learning [5, 33, 42] has emerged as a dominant solution. Our method instead leverages LLMs to generate contextually relevant samples while retaining collaborative signals, allowing PAnDA to handle cold-start more robustly without complex auxiliary data or graph structures.

Large Language Models (LLMs) for Recommendation. LLMs have garnered attention in recommender systems, with various efforts made to model user behavior with LLMs [10, 20, 29]. LLMs have been employed as inference models in various recommendation tasks, such as rating prediction, sequential recommendation, and direct recommendation. Recent work has further explored their potential in addressing cold-start challenges, such as leveraging language-model priors to overcome item cold-start [26], and using keyword-driven retrieval-augmented LLMs to alleviate user cold-start issues [12]. However, most previous approaches primarily utilized LLMs as recommenders [3], focusing on their text-processing capabilities while overlooking the collaborative signals that traditional recommender systems excel at capturing. In this paper, we combine LLM-based data augmentation [30, 43] with traditional data augmentation methods based on collaborative signals, combining both at two levels to achieve preference-aligned augmentation and improve the performance of downstream recommender models.

Data Augmentation for Recommendation. Data augmentation has been a long-standing research focus in recommender systems. Common augmentation operations include permutation, deletion, swapping, insertion, and duplication [17], as well as more recent strategies such as counterfactual reasoning [39] and contrastive learning [27]. Despite these efforts, the quality of augmented data remains an open problem, particularly in sparse or cold-start settings. In this work, we revisit insertion and deletion operations from the perspective of user preference alignment and propose PAnDA tailored for the cold-start scenario.

6 Conclusion

In this paper, we addressed user cold-start recommendation via data augmentation. We analyzed limitations of existing LLM-based methods and identified two key properties of high-quality augmentation: preference alignment and downstream-model awareness. Based on this, we proposed PAnDA, a novel LLM-powered framework that iteratively integrates textual and collaborative signals at both interaction and representation levels, while adaptively filtering inconsistent samples during training. Extensive experiments on three real-world datasets confirmed the benefits of generating preference-aligned and downstream-model-aware augmented data for recommendation tasks. For future work, we plan to enhance scalability and efficiency of the bi-level optimization and conduct more rigorous validations of alignment with true user preferences.

Acknowledgments

This work was supported by the Heilongjiang Key R&D Program of China under Grant No.GA23A915, Australian Research Council (ARC) under Grant No. DP200102611, Hong Kong Research Grants Council's Research Impact Fund (No.R1015-23), Collaborative Research Fund (No.C1043-24GF), General Research Fund (No.11218325), Institute of Digital Medicine of City University of Hong Kong (No.9229503), Huawei (Huawei Innovation Research Program), Tencent (CCF-Tencent Open Fund, Tencent Rhino-Bird Focused Research Program), Alibaba (CCF-Alimama Tech Kangaroo Fund No. 2024002), Ant Group (CCF-Ant Research Fund), Didi (CCF-Didi Gaia Scholars Research Fund), Kuaishou, and Bytedance.

7 GenAI Disclosure Statement

GenAI tools were actively used as part of the research methodology in this work. Specifically, LLMs were employed to generate augmented training samples for cold-start users within the proposed PAnDA framework. These augmented interactions were integrated into the data augmentation pipeline under the authors' full supervision. In addition, LLMs were also used to assist with minor editing and wording improvements in the manuscript. All AI-generated content was carefully reviewed and validated by the authors to ensure accuracy, relevance, and alignment with the research goals.

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